1 Find the value of $16^{-\frac{3}{2}}$

4^{-3}

64^{-1}

$\frac{1}{64}$

$\frac{1}{64}$

(Total for Question 1 is 2 marks)

---

2 Make x the subject of the formula $3x + a = b(x + 5)$

$3x + a = bx + 5b$

$3x - bx = 5b - a$

$x(3 - b) = 5b - a$

$x = \frac{5b - a}{3 - b}$ $\left[\text{or } x = \frac{a - 5b}{b - 3}\right]$

$x = \frac{5b - a}{3 - b}$

(Total for Question 2 is 3 marks)

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3 Expand and Simplify $(x + 5)(x - 2)^2$

$(x + 5)(x - 2)(x - 2)$

$(x^2 - 2x + 5x - 10)(x - 2)$

$(x^2 + 3x - 10)(x - 2)$

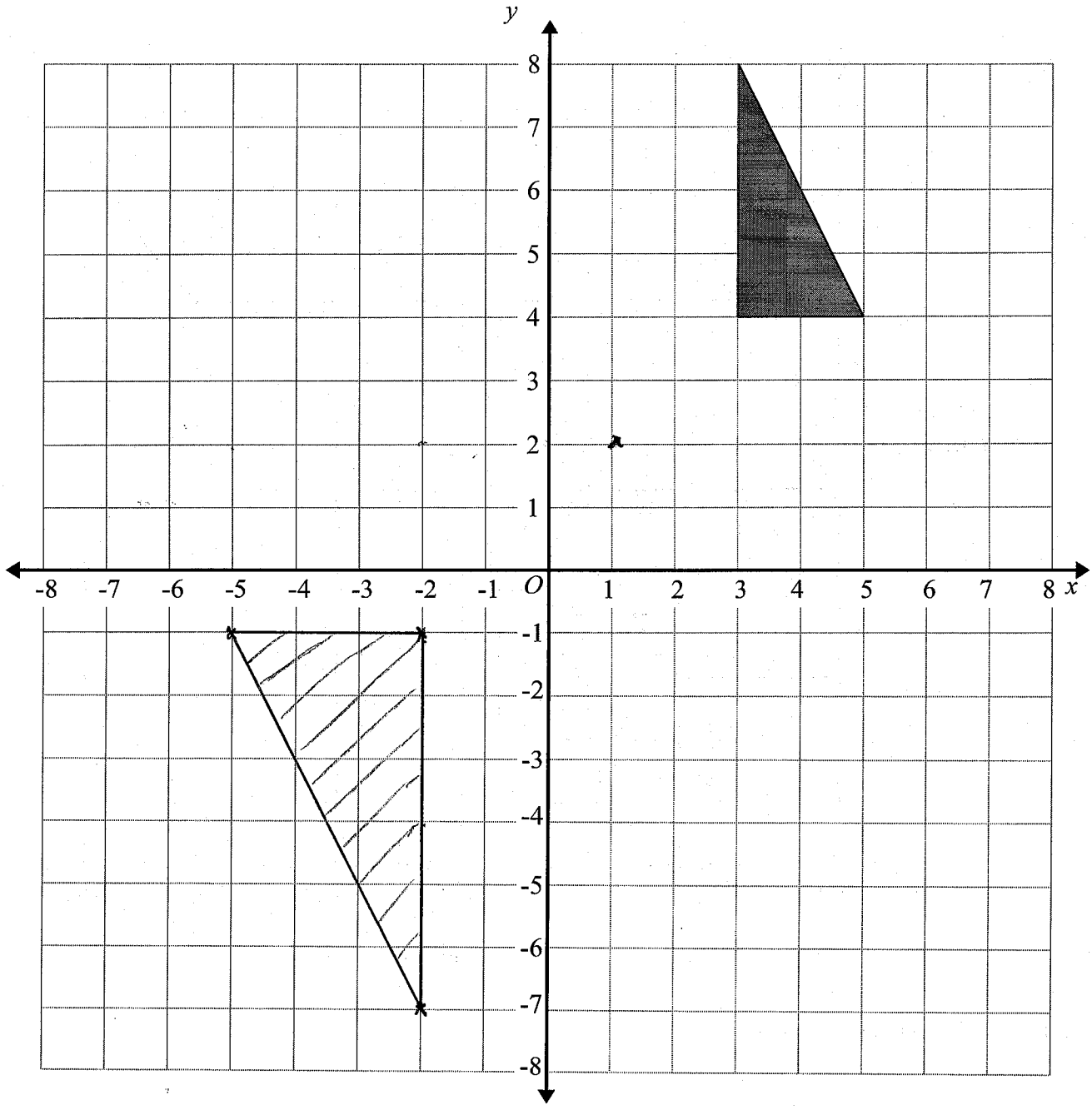
$x^3 - 2x^2 + 3x^2 - 6x - 10x + 20$

$x^3 + x^2 - 16x + 20$

$x^3 + x^2 - 16x + 20$

(Total for Question 3 is 3 marks)

4



On the grid, enlarge the triangle by scale factor -1.5. centre (1, 2).

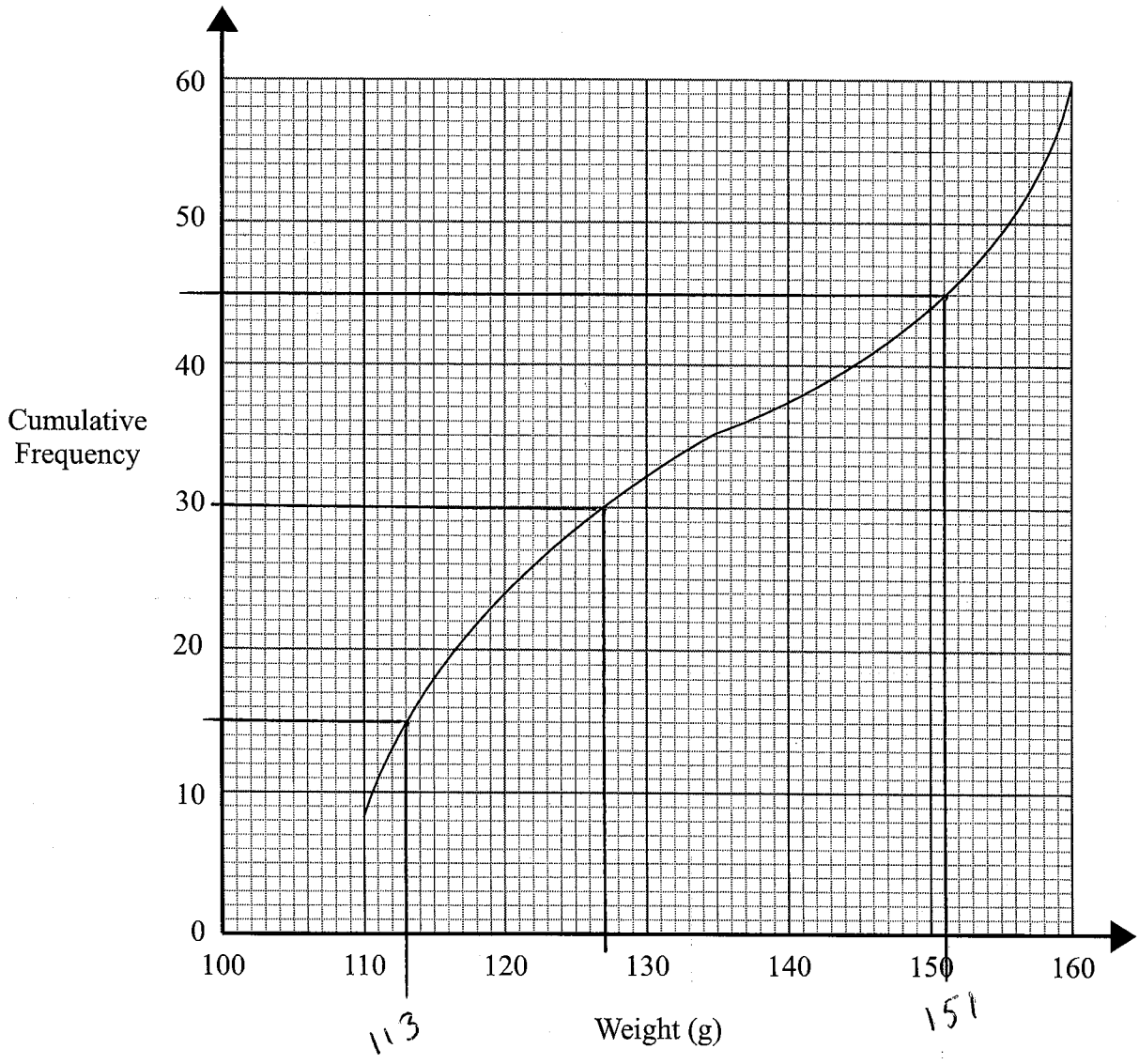
(Total for Question 4 is 2 marks)

$\begin{pmatrix} 2 \\ 2 \end{pmatrix} \times -1.5 = \begin{pmatrix} -3 \\ -3 \end{pmatrix}$

$\begin{pmatrix} 4 \\ 2 \end{pmatrix} \times -1.5 = \begin{pmatrix} -6 \\ -3 \end{pmatrix}$

$\begin{pmatrix} 2 \\ 6 \end{pmatrix} \times -1.5 = \begin{pmatrix} -3 \\ -9 \end{pmatrix}$

5



(a) Find the median weight.

127 g
(1)

(b) Find the inter quartile range.

$$\begin{array}{r} 151 \\ -113 \\ \hline 38 \end{array}$$

38 g
(2)

(Total for Question 5 is 3 marks)

6 Simplify fully $\dfrac{7+2\sqrt{5}}{2+\sqrt{5}}$

$$\frac{(7+2\sqrt{5})(2-\sqrt{5})}{(2+\sqrt{5})(2-\sqrt{5})}$$

$$\frac{14-7\sqrt{5}+4\sqrt{5}-2(5)}{4-2\sqrt{5}+2\sqrt{5}-5}$$

$$\frac{14-3\sqrt{5}-10}{-1}$$

$$\frac{4-3\sqrt{5}}{-1} = -4+3\sqrt{5}$$

$-4+3\sqrt{5}$

(Total for Question 6 is 3 marks)

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7 Prove algebraically that the recurring decimal $0.3\dot{1}\dot{8}$ can be written as $\dfrac{7}{22}$

$0.3\dot{1}\dot{8} = x$

$3.\dot{1}\dot{8} = 10x$

$318.\dot{1}\dot{8} = 1000x$

$315 = 990x$

$x = \dfrac{315}{990}$

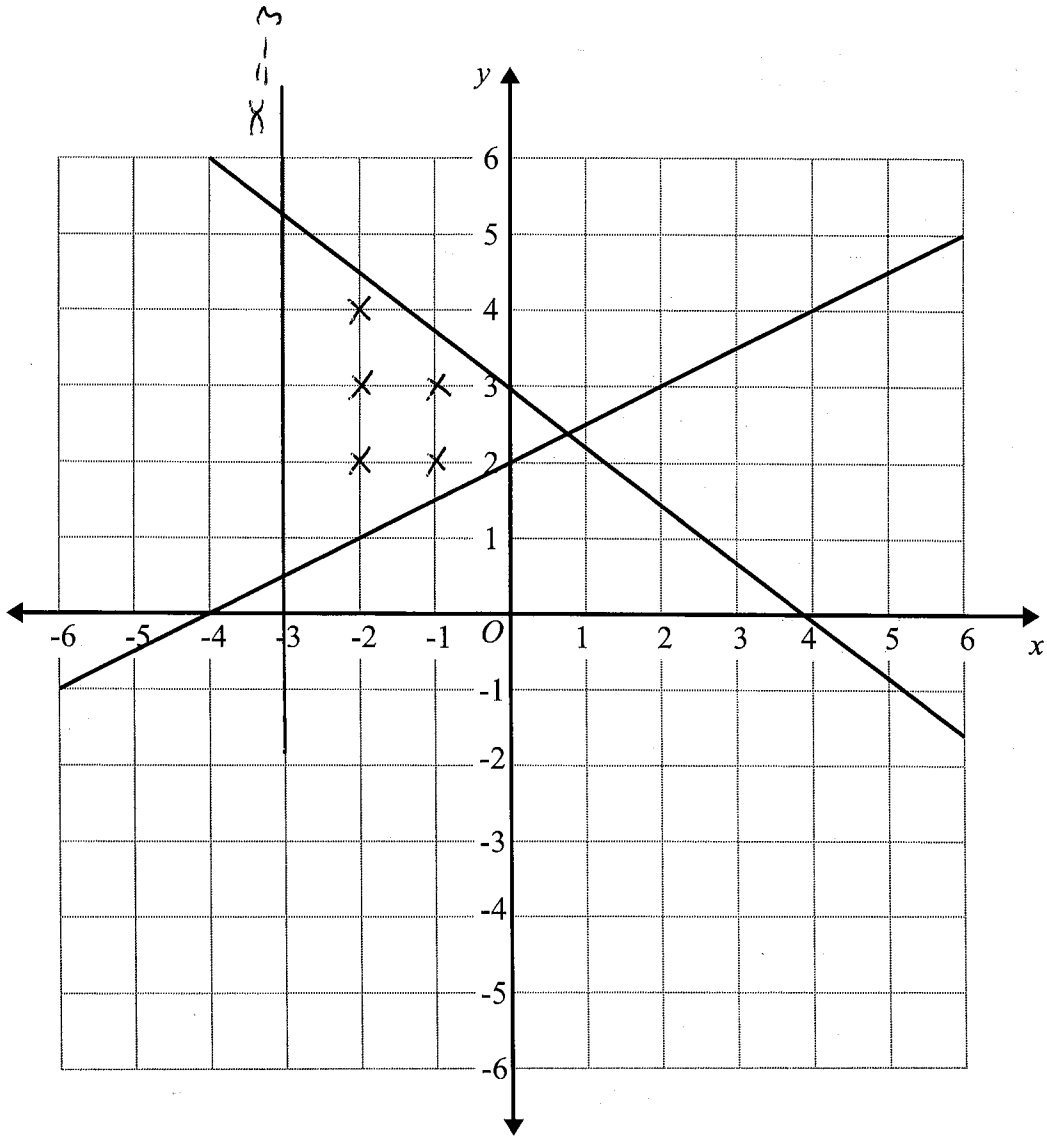
$x = \dfrac{315}{990}$

$= \dfrac{105}{330}$

$= \dfrac{21}{66} = \dfrac{7}{22}$

(Total for Question 7 is 2 marks)

8 The graphs of the straight lines with equations $2y = x + 4$ and $4y + 3x = 12$ have been drawn on the grid.



x and y are both integers.

Mark with a cross (×) all of the points that satisfies all the inequalities

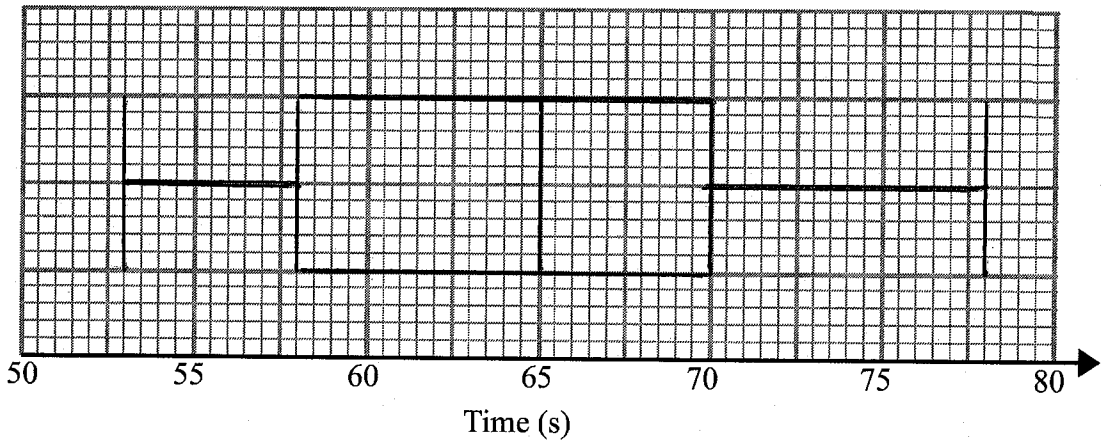
$2y > x + 4$ $\qquad$ $4y + 3x < 12$ $\qquad$ $x > -3$

(Total for Question 8 is 2 marks)

9 The times, in seconds, of 15 students running a race are recorded below.

53 54 57 58 58 60 61 65 67 67 68 70 71 71 78

Draw a box plot for this information.



(Total for Question 9 is 2 marks)

$x_1 \quad y_1 \quad x_2 \quad y_2$

10 Line A passes through the points (2, -6) and (12, -1)
Find the equation of the line perpendicular to A that passes through (2,5)

$m = \dfrac{y_2 - y_1}{x_2 - x_1}$

$= \dfrac{-1 - -6}{12 - 2}$

$= \dfrac{5}{10}$

$= \dfrac{1}{2}$

perpendicular gradient $= -2$

$y = -2x + c \quad (2, 5)$

$5 = -2(2) + c$

$5 = -4 + c$

$c = 9$

$y = -2x + 9$

(Total for Question 10 is 2 marks)